

Devasmito Das

Maître de conférences in Industrial Engineering | IUT de Béziers, Université de Montpellier
Researcher, DEXTER Team, LIRMM UMR CNRS 5506 | Soft Robotics – Nonlinear and Intelligent Control
Fractional-order systems, nonlinear dynamics, model predictive control, robotics and machine learning
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Profile

Maître de conférences in Industrial Engineering at IUT de Béziers, Université de Montpellier, and researcher within the **DEXTER team at LIRMM UMR CNRS 5506**. My research combines nonlinear dynamics, advanced control, fractional-order modelling and intelligent methods, with a growing focus on soft robotics and experimentally relevant robotic systems.

Core Expertise

- Soft robotics and robotic systems
- Nonlinear dynamical systems
- Fractional-order systems
- Model Predictive Control / NMPC
- Intelligent and learning-based control
- Chaos and synchronization
- Stability and nonlinear control
- State estimation and EKF
- Physics-informed learning
- Coupled oscillator networks

Technical Skills

Programming

MATLAB Python C++

Control / Simulation

Simulink CasADi YALMIP

Scientific Tools

LaTeX Git Docker ROS 2

Data / Computing

NumPy Pandas Matplotlib

Languages

English	Fluent
French	Fluent
Bengali	Native
Hindi	Native
Spanish	Basic reading
Japanese	Basic speaking

Research Keywords

Soft robotics • Nonlinear control • Fractional calculus • MPC • Machine learning • Chaos • Synchronization • State estimation • Complex networks

Research Identity

My work aims to develop robust, predictive and intelligent control methods for complex systems exhibiting nonlinearities, memory, uncertainty and constraints.

Current Academic Position

Maître de conférences in Industrial Engineering

2026–Present

IUT de Béziers, Université de Montpellier, France

Teaching and student supervision in programming, automatic control, modelling and industrial/robotic applications. Research activities are conducted within the **DEXTER team at LIRMM UMR CNRS 5506**, with a focus on soft robotics, nonlinear and predictive control, fractional-order modelling and intelligent computational methods.

Education

PhD in Automation, Control and Robotics

2022–2025

École Centrale de Nantes / LS2N, France & IIT Madras, India

Thesis: *Coupled Network Fractional Nonlinear Oscillators*. Supervisors: J.-J. Loiseau, I. Taralova and M. Pandey.

Master's Degree in Control and Robotics – Advanced Robotics

2019–2021

École Centrale de Nantes, France

Robotics, control, perception, modelling and intelligent systems.

Bachelor of Technology in Electrical Engineering

2014–2018

Meghnad Saha Institute of Technology, India

Electrical engineering, control, machines, power systems, electronics and applied mathematics.

Research Experience

Soft robotics and advanced control

- Research within the DEXTER team at LIRMM on modelling, control and optimization of complex robotic systems, particularly soft robots.
- Development of nonlinear and predictive control methods for systems with uncertainty, memory, constraints and nonlinear actuation.

Fractional-order nonlinear dynamics

- Modelling and numerical analysis of fractional Rössler, Duffing, Lorenz and coupled oscillator systems.
- Analysis of chaos, synchronization, bifurcations and stability using fractional numerical schemes.

Control, estimation and intelligent methods

- Developed time-delay control, feedback linearization, sliding-mode control, backstepping, MPC and fractional MPC.
- Integrated reinforcement learning, physics-informed learning, self-supervised approaches and extended Kalman filtering.

Current Research Perspective

My current work extends my doctoral expertise in fractional-order nonlinear dynamics toward experimentally oriented robotic systems. Within DEXTER, I am particularly interested in nonlinear and predictive control for soft robots, including systems affected by uncertainty, actuator nonlinearities, memory effects and physical constraints.

International Research Collaboration

Franco–Bulgarian Research Collaboration / PHC RILA

2024–2025

- Previous experience within a bilateral PHC RILA project involving École Centrale de Nantes / LS2N and the Technical University of Sofia, including research mobility at TU Sofia in June 2025.
- Presented research at IEEE ICARAI 2025 in Sozopol and collaborated with researchers from TU Sofia and École Centrale de Nantes.
- Contributed to multiple joint works with T. Slavov and S. Filipova-Petrakieva, providing an established basis for further Franco–Bulgarian collaboration.

Teaching Experience

Université de Montpellier

IUT de Béziers 2026–Present

- Maître de conférences in Industrial Engineering.
- Programming, control, modelling and robotic/industrial applications.
- Tutorials, practical sessions and student projects.

École Centrale de Nantes 2022–2024

- Control and Observation of Systems.
- Professional engineering project supervision.

IIT Madras

2024–2025

- Design of Machine Elements.
- Theory of Vibrations.
- Rotor Dynamics laboratory.

Academic Service

- Reviewer, IFAC World Congress 2026
- Reviewer, iAIMS 2026
- Reviewer, ICAISF 2026
- Reviewer, ICECET 2026
- Reviewer, IWACCE 2025
- Reviewer, TELKOMNIKA

Qualification

CNU Qualification – Section 61 2026

Qualified for Maître de conférences positions in Automatic Control, Signal Processing and related engineering disciplines in France.

Scientific Outreach

- Scientific mediation through the DECLICS programme in Nantes.
- Presented nonlinear dynamics, AI and neuromorphic computing to high-school students.

Current Research Focus

- Soft robotics
- Nonlinear MPC
- Memory-dependent modelling
- Fractional-order dynamics
- Intelligent control
- Physics-informed learning
- Robotic systems under constraints

Collaboration Profile

Experienced in international research involving France, India and Bulgaria, combining analytical modelling, numerical simulation, control design and collaborative scientific publication.

Selected Publications

- **Das, D.**, Taralova, I., Loiseau, J.-J. (2024). Time-delay feedback control of fractional chaotic Rössler oscillator. *IFAC-PapersOnLine*, 58(5), 90–95.
- **Das, D.**, Taralova, I., Loiseau, J.-J. (2026). Analysis of fractional MPC nonlinear control applied to fractional Rössler oscillator. In *Proceedings of the 17th Chaotic Modeling and Simulation International Conference*, Springer Proceedings in Complexity.
- **Das, D.**, Taralova, I., Loiseau, J.-J., Filipova-Petrakieva, S. (2025). Fractional model predictive control of fractional chaotic Rössler oscillator. *IFAC-PapersOnLine*, 59(11), 156–161.
- **Das, D.** et al. (2025). Frequency-domain identification of 1-DoF and 3-DoF fractional-order Duffing systems using Grünwald–Letnikov characterization. *Fractal and Fractional*, 9(9), 581.
- **Das, D.** et al. (2025). Active control via nonlinear feedback linearization for coupled fractional-order Rössler oscillators. *IEEE ICARAI 2025*, Sozopol, Bulgaria.
- **Das, D.** et al. (2025). State estimation using extended Kalman filter for fractional model predictive control of fractional chaotic Rössler oscillator. *CoDIT 2025*, Split, Croatia.
- **Das, D.** et al. (2025). Reinforcement learning-based PID control of Rössler oscillator. *ICECET 2025*, Paris, France.
- **Das, D.** et al. (2025). Reinforcement learning for PID control of fractional Rössler oscillator. *IEEE International Symposium on Systems Engineering*, Paris, France.
- **Das, D.** et al. (2025). Fractional mixed-integer model predictive control of fractional Rössler oscillator. *IFAC ICFDA 2025*, Alger, Algeria.
- **Das, D.** et al. (2025). Fractional-order coupled oscillators for synchronization-based gait control in hexapod robots. *CHAOS 2025*, Springer Nature, in press.
- **Das, D.**, Taralova, I., Loiseau, J.-J. (2026). Adaptive fuzzy sliding mode control for fractional-order Rössler system using Grünwald–Letnikov fractional integrals. *CHAOS 2026*. Accepted.
- **Das, D.** et al. (2026). Adaptive backstepping control of incommensurate fractional-order Rössler oscillator with derivative orders greater than unity. *IFAC World Congress 2026*. Accepted.
- **Das, D.** (2026). Game-theoretic nonlinear model predictive control for chaos suppression and orbital stabilization in coupled fractional-order Duffing oscillators. *Nonlinear Science*, 100141.
- **Das, D.** (2026). A fractional-order hyperchaotic framework for dynamic affine S-box parameterization and bidirectional diffusion with neural nonlinear mixing. *Nonlinear Science*, 100162.

Current and Emerging Research Directions

- Nonlinear and predictive control of soft robotic systems.
- Modelling of robotic systems with memory, uncertainty and nonlinear actuator dynamics.
- Fractional-order modelling of viscoelastic and hereditary systems.
- Physics-informed and machine-learning-assisted control.
- Intelligent state estimation for nonlinear dynamical systems.
- Synchronization and stability of complex oscillator networks.
- Koopman and graph-based representations for nonlinear control.

Research Perspective

My current research programme connects my background in fractional-order nonlinear dynamics, advanced control and state estimation with experimentally oriented problems in robotics. Within the DEXTER team at LIRMM, I aim to develop modelling, estimation and predictive-control methods for soft robotic systems where compliance, nonlinear actuation, memory and physical constraints play a central role.

Summary

As a Maître de conférences at Université de Montpellier and researcher within the DEXTER team at LIRMM, my objective is to develop robust and intelligent control methods for nonlinear and complex robotic systems while continuing fundamental research on fractional-order dynamics, predictive control, machine learning and complex networks.